

F24-141-D-DermaScan

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Chapter 1

Introduction

The following Software Requirement Specification (SRS) document outlines the design, architecture, functional and non-functional requirements of our application (DermaScan-AI Powered Skin Disease Detector). This project leverages cutting-edge machine learning model for efficient skin disease detection in addition to provision of other integrated features including LLM-powered chatbot and doctor recommendations. This document ensures the clear understanding of the project vision, structure, features, constraints and project goals.

1.1 Problem Statement

Lack of dermatologists and healthcare services, particularly in remote or underserved areas results in delayed treatment of skin diseases. Additionally, due to the lack of these services, many individuals are in worse conditions of serious diseases. Also visiting doctors for regular treatment is costly and time-consuming. Our mobile app will provide an AI-driven system to detect skin diseases and also suggest a doctor if the individual is going to worse conditions just with our mobile application. By offering early detection, progress tracking, and Dermatologist's recommendation our app will help users to manage skin health care and also reduce the burden on health care systems.

1.2 Scope

The proposed project aims to develop a mobile phone application that enables early detection of skin diseases. The team will begin by collecting data from publicly available datasets and extract relevant data to the subcontinent. An important part of the project is to explore available CNN architectures that can provide faster and accurate detection of

skin diseases. Architectures that use less time and compute resources will be preferred as it is for a mobile phone application. The main focus will be on having a higher recall by the model. The diseases will be targeting are as follows, but more could be added if data is available:

1. Psoriasis
2. Acne
3. Eczema
4. Vitiligo
5. Skin Cancer

Another important feature that will be developed during the project will be an LLM powered medical chatbot. The chatbot will be posed as a virtual consultation taking symptoms related to detected disease to improve the accuracy of CNN based detection. The chatbot will further assist by providing support and guidance to the patient.

The last part of the project is to provide users access to quality healthcare. This will be achieved by providing users relevant doctor information in nearby vicinity so the user is able to make an informed decision. Ultimately the goal is for providing end to end access to user where they are able to identify their disease, gain insights into it and finally be able to access quality healthcare nearby.

1.3 Modules

Following are the detailed modules of our system.

1.3.1 Module 1: Patient Profile

This module focuses on the development of the general user profile interfaces which include the following features.

1. Personal profile management.
2. Medical records management.
3. Disease symptoms logging for patient diseases.
4. User Feedback submission

1.3.2 Module 2: Skin Disease Detection

This module focuses on skin disease detection using a CNN model and is trained using various image datasets. This module includes the integration of the camera to capture user images, processing them, and predicting outcomes using the CNN model. This module also includes generation of diagnosis results based on user input.

Following steps will be followed to train the CNN model for skin disease detection.

1. Image capturing using mobile camera.
2. Skin-disease detection using CNN model.
3. Diagnosis results' report generation.
4. Tracking progress of disease improvement.
5. Doctor recommendations based on disease and region.
** Please note that this module leverages a CNN model, trained on ISIC, HAM10000, and Kaggle datasets, for disease detection.*

1.3.3 Module 3: LLM powered Chatbot

The following module includes the integration of an LLM powered chatbot for improved disease detection in addition to providing general information and guidance related to skin diseases relevant to a user.

Following are the features included in this module.

1. Disease detection using symptoms logged in chat bot.
2. Disease-relevant guidance and information generation through Chat bot.
** Please note that this module leverages a LLM model for disease detection and information generation.*

1.3.4 Module 4: Doctor Profile

This module focuses on development of gateway for doctors to access and register on the app. This module ensures communication between patients and relevant doctors.

1. Doctor profile management
2. Geo-location based recommendations.

1.4 User Classes and Characteristics

Following is the list of the user classes that will use our app.

1. User (Patients)
2. Dermatologist (Healthcare Provider)
3. Researcher (Medical or AI/ML Researcher)

User class	Description
Patients	user classes that want to efficiently diagnose a possible skin disease or get relevant information without leaving the comfort of their homes.
Dermatologists (Healthcare Provider)	healthcare professionals who want to offer their services to patients on the app in addition to connecting with patients for future appointments or meetings.
Researcher (Medical or AI/ML Researcher)	Researchers either in the medical or AI/ML field, interested in analyzing the algorithms used to train models used in the app to evaluate the AI effectiveness and potentially improve the algorithms.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

The use case diagram below provides the visual representation of actors and their interaction to different use cases within the DermaScan app.

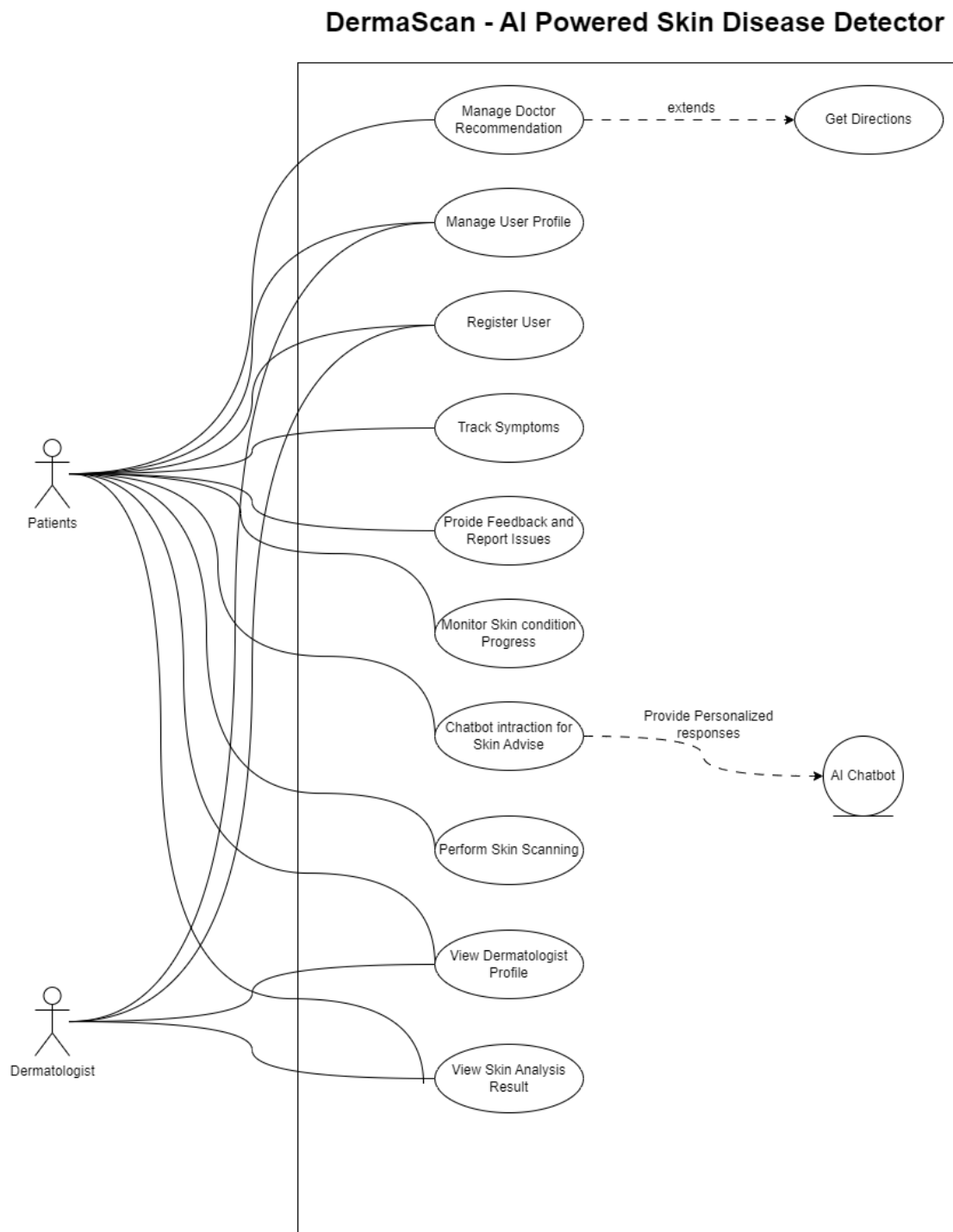


Figure 2.1: Use Case Diagram

Following are the detailed use cases of the project.

2.1.1 Use Case 1 - Register User

Use Case ID	UC 01
Use Case Name	Register User
Scope	DermaScan App
Primary Actor	• Patients • Dermatologists
Description	All new users must register on the app to access its features.
Stakeholders and Interests	All user classes who want to access the app features.
Pre-Conditions	• Active internet connection. • The user has downloaded the DermaScan App.
Post-Conditions	• User account is successfully created and registered. • User can access the app after login.
Success Guarantee	User details are saved in the database.
Alternatives	• User cancels the registration process.
Extensions	If invalid credentials are entered, account registration fails and appropriate error message is prompted.
Exceptions	User account with entered email already exists.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Frequently, as new users will join the app.

Miscellaneous	None
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Table 2.1: Use Case 1: Register User

Step	User Action	System Response
1	User opens the DermaScan app and clicks on the "Register" button.	The system navigates the user to the registration page.
2	User enters their credentials including name, email and password.	The system processes and temporarily stores the entered details.
3	User presses the create account button.	The system validates user credentials and creates a new account.
4	Registration successful.	The system redirects the user to the login page after successful registration.

Table 2.2: Event Response Use Case 1: Register User

2.1.2 Use Case 2 - User Login

Use Case ID	UC 02
Use Case Name	User Login
Scope	DermaScan App
Primary Actor	• Patients • Dermatologists
Description	Registered users log in to access the app's features.
Stakeholders and Interests	Users who want to access the app.
Pre-Conditions	• The user is already registered. • The user has active internet connection.
Post-Conditions	• User is successfully logged in.

Success Guarantee	User credentials are successfully authenticated against the stored data.
Alternatives	• The user navigates to the Sign up page. • The user navigates to the forget password tab.
Extensions	In case of invalid credentials, an error message is displayed.
Exceptions	The user isn't registered on the app.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Most Frequent, as users must log in to access the app features.
Miscellaneous	None

Table 2.3: Use Case 2: User Login

Step	User Action	System Response
1	User opens the DermaScan app and clicks on the "Login" button.	The system loads the login page.
2	User enters credentials including email and password and clicks on the Sign In button.	user is authenticated and redirected to the Home screen.

Table 2.4: Event Response Use Case 2: User Login

2.1.3 Use Case 3 - Manage User Profile

Use Case ID	UC 03
Use Case Name	Manage User Profile
Scope	DermaScan App
Primary Actor	• Patients • Dermatologists

Description	Users can view and update their personal profile details.
Stakeholders and Interests	Users who want to manage their profile information.
Pre-Conditions	• The user is logged in.
Post-Conditions	• changes in profile data are reflected in the app.
Success Guarantee	Changes to the profile are saved in the database.
Alternatives	• The user can cancel changes and exit the profile section.
Extensions	In case of an error, an appropriate message is displayed.
Exceptions	System fails to save changes.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Occasional, users update profiles occasionally.
Miscellaneous	None

Table 2.5: Use Case 3: Manage User Profile

Step	User Action	System Response
1	User navigates to the profile section in the app.	The system loads the profile information.
2	User edits their profile information and submits the changes.	The system updates the profile in the database and confirms the changes.
4	User receives a confirmation message.	The system displays a message indicating the profile has been successfully updated.

Table 2.6: Event Response Use Case 3: Manage User Profile

2.1.4 Use Case 4 - Perform Skin Scanning

Use Case ID	UC 04
Use Case Name	Perform Skin Scanning
Scope	DermaScan App
Primary Actor	• Patients
Description	Users upload a picture of their skin to detect potential skin conditions.
Stakeholders and Interests	patients and seeking diagnostics for skin conditions.
Pre-Conditions	• The user is logged in. • The device camera is functioning properly.
Post-Conditions	• The image is uploaded for analysis. • The user receives confirmation of upload.
Success Guarantee	The image is successfully processed by the system and forwarded for scan.
Alternatives	• The user can cancel the scan process at any time.
Extensions	In case of upload failure, an error message is displayed.
Exceptions	None
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Highly frequent, as it is the main feature and expects high traffic
Miscellaneous	None

Table 2.7: Use Case 4: Perform Skin Scanning

Step	User Action	System Response
1	User navigates to the home page.	The system loads content of the home page.
2	User selects the "Scan Skin" option in the app.	The system opens the camera interface for capturing an image.
3	User captures a picture of the affected skin area.	The system verifies the image quality and prompts for upload.
4	User confirms the upload.	The system confirms the upload and is forwarded for diagnosis.

Table 2.8: Event Response Use Case 4: Perform Skin Scanning

2.1.5 Use Case 5 - View Skin Analysis Results

Use Case ID	UC 05
Use Case Name	View Skin Analysis Results
Scope	DermaScan App
Primary Actor	• Patients
Description	Users can view the analysis results of their uploaded samples.
Stakeholders and Interests	Users seeking to understand their skin conditions.
Pre-Conditions	• The user has uploaded a skin image for analysis. • The analysis process is complete.
Post-Conditions	• The user can view the analysis results.
Success Guarantee	The analysis results are successfully generated and displayed.

Alternatives	• None
Extensions	In case the analysis fails, an error message is displayed.
Exceptions	The analysis results are not available due to a processing error.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Frequent, after every skin scan.
Miscellaneous	None

Table 2.9: Use Case 5: View Skin Analysis Results

Step	User Action	System Response
1	User navigates to the "View Results" section.	The system displays a loading screen while retrieving results.
2	User views the analysis results.	The system displays detailed results, including diagnosis and severity.
3	User saves the results	The system ensures the download of the analysis results
4	User shares the analysis results with dermatologists.	The system ensures the download of the analysis results

Table 2.10: Event Response Use Case 5: View Skin Analysis Results

2.1.6 Use Case 6 - Track Symptoms

Use Case ID	UC 06
Use Case Name	Track Symptoms
Scope	DermaScan App
Primary Actor	• Patients
Description	Users want to monitor their skin-related symptoms over time.

Stakeholders and Interests	Users seeking to track changes in their skin conditions.
Pre-Conditions	• The user is logged in. • The user has forwarded test samples for scan.
Post-Conditions	• The symptoms are successfully recorded and can be reviewed later.
Success Guarantee	User symptoms are saved and available for viewing.
Alternatives	None
Extensions	In case of invalid input, appropriate error message is displayed.
Exceptions	User input cannot be processed due to a system error.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Regularly, as users monitor their symptoms over time.
Miscellaneous	None

Table 2.11: Use Case 6: Track Symptoms

Step	User Action	System Response
1	User selects the "Track Symptoms" option in the app.	The system redirects the user to the "Symptoms tracking" screen.
2	User inputs details about their current symptoms (e.g., severity, duration) and clicks on the "submit" button.	The system validates the input and saves the details to user profile.
4	User views their symptom history.	The system displays a summary of logged symptoms and trends over time.

Table 2.12: Event Response Use Case 6: Track Symptoms

2.1.7 Use Case 7 - Chatbot Interaction for Skin Advice

Use Case ID	UC 07
Use Case Name	Chatbot Interaction for Skin Advice
Scope	DermaScan App
Primary Actor	• Patients
Description	Users interacts with a chatbot to receive skin care advice and information.
Stakeholders and Interests	Users seeking support and guidance for skin-related queries.
Pre-Conditions	• The user is logged in.
Post-Conditions	• The chat-bot generates relevant information corresponding to user input.
Success Guarantee	User queries are accurately answered by the chat-bot.
Alternatives	None
Extensions	In case of insufficient or inappropriate input, an error message is displayed.
Exceptions	System fails to connect to the chatbot service.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Frequent, as users may have multiple queries.
Miscellaneous	None

Table 2.13: Use Case 7: Chatbot Interaction for Skin Advice

Step	User Action	System Response
1	User selects the "Chatbot" option in the app.	The system redirects the user to the chatbot interface.
2	User types a question or concern regarding their skin condition.	The system processes the query and generates corresponding output.

Table 2.14: Event Response Use Case 7: Chatbot Interaction
for Skin Advice

2.1.8 Use Case 8 - Monitor Skin Condition Progress

Use Case ID	UC 09
Use Case Name	Monitor Skin Condition Progress
Scope	DermaScan App
Primary Actor	• Patients
Description	Users can track and review changes in their skin condition over time.
Stakeholders and Interests	Users who want to monitor their skin health and treatment effectiveness.
Pre-Conditions	• The user has previously logged symptoms or uploaded skin images.
Post-Conditions	• Treatment history is successfully displayed
Success Guarantee	The system successfully retrieves and displays the logged data.
Alternatives	None
Extensions	In case of no recorded data, an appropriate message is displayed.
Exceptions	System fails to retrieve data due to a processing error.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Regularly, as users track their skin conditions over time.
Miscellaneous	None

Table 2.15: Use Case 8: Monitor Skin Condition Progress

Step	User Action	System Response
1	User selects the "Monitor Progress" option in the app.	The system displays a history of logged symptoms and images.
2	User reviews the changes in their skin condition over time.	The system presents graphical representations (e.g., charts) of symptom severity and frequency.

Table 2.16: Event Response Use Case 8: Monitor Skin Condition Progress

2.1.9 Use Case 9 - Manage Doctor Recommendations

Use Case ID	UC 9
Use Case Name	Manage Doctor Recommendations
Scope	DermaScan App
Primary Actor	• Patients
Description	Users can view and manage recommendations for dermatologists based on their skin conditions.
Stakeholders and Interests	Users seeking professional help for skin conditions.
Pre-Conditions	• The user has completed a skin scan.
Post-Conditions	• The user can view, save, or discard doctor recommendations.
Success Guarantee	Recommendations are accurately retrieved and displayed.
Alternatives	None
Extensions	In case no recommendations are available, an appropriate message is displayed.
Exceptions	System fails to retrieve recommendations due to a processing error.

Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Frequent, as users may need to consult doctors based on their condition.
Miscellaneous	None

Table 2.17: Use Case 9: Manage Doctor Recommendations

Step	User Action	System Response
1	User navigates to the "Doctor Recommendations" section in the app.	The system retrieves and displays relevant doctor recommendations list.
2	User views and selects a doctor for more details.	The system displays additional information about the selected doctor, including ratings and reviews.
4	User can save the recommendation or get directions using the map feature.	The system confirms the action and either saves the recommendation or opens the map for directions.

Table 2.18: Event Response Use Case 9: Manage Doctor Recommendations

2.1.10 Use Case 10 - Provide Feedback and Report Issues

Use Case ID	UC 10
Use Case Name	Provide Feedback and Report Issues
Scope	DermaScan App
Primary Actor	• Patients
Description	Users can provide relevant feedback about the app.
Stakeholders and Interests	Users who wish to improve their experience and report problems.
Pre-Conditions	• User is logged In.

Post-Conditions	• The feedback is successfully submitted.
Success Guarantee	User feedback is stored and can be reviewed by the development team.
Alternatives	None
Extensions	In case of submission failure, an error message is displayed.
Exceptions	System fails to log the feedback due to a processing error.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Occasional, as users provide feedback or report issues as needed.
Miscellaneous	None

Table 2.19: Use Case 10: Provide Feedback and Report Issues

Step	User Action	System Response
1	User navigates to the "Feedback" section in the app.	The system redirects the user to the feedback form.
2	User fills in the feedback form and clicks on the "submit" button.	The system prompts for confirmation to submit the feedback.
3	User confirms the submission.	The system saves the feedback to the database and notifies the user.

Table 2.20: Event Response Use Case 10: Provide Feedback and Report Issues

2.1.11 Use Case 11 - View Dermatologist Profiles

Use Case ID	UC 11
Use Case Name	View Dermatologist Profiles
Scope	DermaScan App

Primary Actor	• Patients
Description	Users can view profiles of dermatologists for potential consultation.
Stakeholders and Interests	Users seeking qualified dermatologists.
Pre-Conditions	• The user is logged in.
Post-Conditions	• dermatologists details are successfully displayed.
Success Guarantee	The system accurately retrieves and displays dermatologist profiles.
Alternatives	None
Extensions	In case no profiles are found, an appropriate message is displayed.
Exceptions	System fails to retrieve profiles due to a processing error.
Special Requirements	None
Technology and Data Variation	None
Frequency of Occurrence	Occasional, as users look for dermatologists when needed.
Miscellaneous	None

Table 2.21: Use Case 11: View Dermatologist Profiles

Step	User Action	System Response
1	User selects the "Find a Dermatologist" option in the app.	The system retrieves a list of available dermatologists.
2	User selects a dermatologist for more details.	The system displays a detailed profile, including qualifications, contact information, and patient reviews.
3	User can save the profile or get directions using the map feature.	The system confirms the action and either saves the profile or opens the map for directions.

Table 2.22: Event Response Use Case 11: View Dermatologist Profiles

2.2 Functional Requirements

Below listed are the functional requirements of each module of our app

2.2.1 Module 1: Patient Profile

Following are the functional requirements of module 1.

1. The system shall allow users to login/register on the app.
2. The system shall allow users to view and update their user profiles.
3. The system shall allow users to maintain their medical records and disease history.
4. The system shall allow users to log their symptoms for a particular disease.
5. The system shall allow users to write and submit feedback reports related to the app's functionality as well as doctors.

2.2.2 Module 2: Skin Disease Detection

Following are the functional requirements for module 2:

1. The system shall allow users to access the phone's camera to capture images for disease detection using the app.
2. The system shall enable users to obtain results of their analysis tests generated by the CNN model.
3. The system shall allow users to save or download the obtained reports for obvious reasons.
4. The system shall provide users with recommended doctors based on disease and geo-location.

2.2.3 Module 3: LLM powered Chatbot

Following are the functional requirements for module 3:

1. The system shall allow users to interact with the chatbot using Text prompts.
2. The system shall ensure improved disease detection by catering to the user prompted symptoms in addition to the disease report generated by the CNN model.
3. The system shall provide users with accurate information and feedback for entered user prompt.

2.2.4 Module 4: Doctor Profile

Following are the functional requirements for module 4:

1. The system shall provide a separate gateway for healthcare professionals.
2. The system shall allow healthcare professionals to view and update their profiles.
3. The system shall enable and ensure connection and communication between patients and doctors.
4. The system shall cater to doctor's geo location when recommending them to a patient.

2.3 Non-Functional Requirements

Following are the non-functional requirements of the system.

2.3.1 Performance

- The system should provide skin disease detection results within 5 seconds after image processing.
- The chatbot should respond to user queries within 5 seconds to ensure smooth interaction.
- The CNN model should process at least 100 image requests per minute.

2.3.2 Security

All user data will be stored in Firebase and protected through Firebase security and hence will require at least months for a possible data breach.

2.3.3 Availability

The system should have an uptime of at least 99.9%.

2.3.4 Usability

The app should have a user-friendly interface in addition to smooth navigation between screens and performance of features.

2.3.5 Reliability

- The CNN model should have an accuracy of 91%.
- The LLM powered chatbot should have an accuracy up to 95%

2.4 Domain Model

The domain model below illustrates the key entities, their attributes, and the relationships between them.

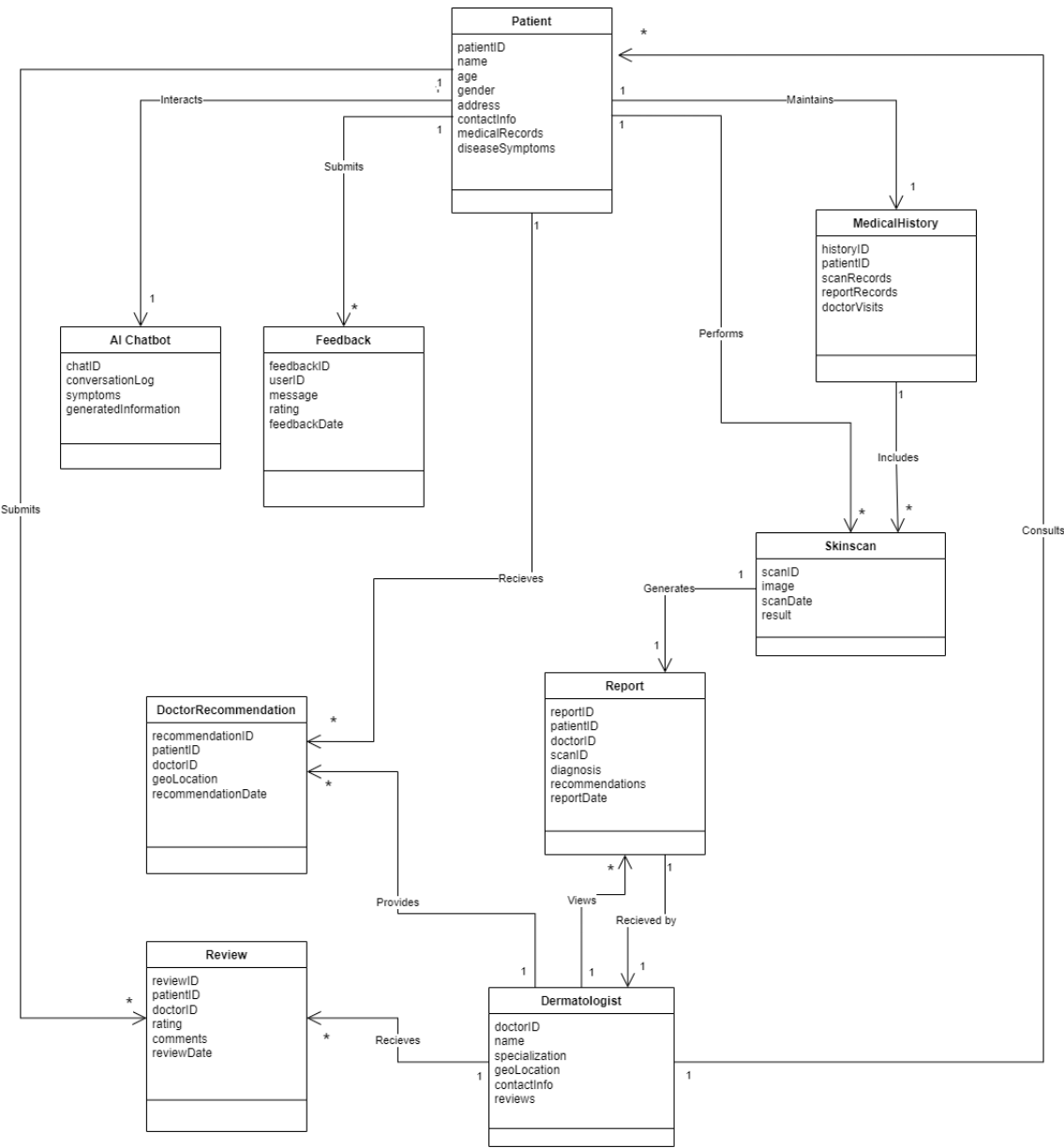


Figure 2.2: Domain Model

Chapter 3

System Overview

DermaScan app is a mobile-based solution designed for the early detection and monitoring of skin diseases efficiently using AI powered image analysis. The app allows patients to upload images of affected areas of skin, which are then analyzed by the CNN model (trained over a large dataset). It also include a LLM powered AI Chatbot for the guidance and symptoms logging and Doctor recommendation system. The overall functionality of the system is to offer real time skin disease detection and a way to consult relevant doctors based on the user region.

3.1 Architectural Design

The DermaScan Skin-Disease and Guidance app is divided into a modular structure to enhance feature integration, maintenance and scalability. The system has 4 main modules including Patient profile, Skin disease detection, LLM powered chatbot and Doctor Profile. Each of the 4 module cater to a different section and functionality of the app and thus promote individuality and independence of app features. However, the modules collaborate with each other to provide the overall functionality of the app.

3.1.1 Modular Relationship

Below mentioned is a brief description of the relationship and collaboration of each module with the rest of the system.

1. **Patient Profile** This module covers the registration and user profile management section of the app and collaborates with the rest of the system by feeding user data including medical records. It provides the user data required for CNN processing in the second module, provides user information for chatbot functionality in the third

module and provides user location for doctor recommendations functionality in the fourth module.

2. **Skin Disease Detection** This module contains the core functionality of the app and collaborates with the first module by fetching user data in addition to users' skin scans for skin-disease detection. It collaborates with the third model by providing the CNN scan results to the chatbot for further processing. This module fetches doctor profiles from the fourth module and recommend it to users based on the "geo-location" obtained from the first module.
3. **LLM powered Chatbot** This module features the LLM powered chatbot and uses logged systems from the first module and scan results from second module to improve the efficiency of the skin-disease detection system. This module has no direct relationship with the doctors' module.
4. **Doctor Profile** This module features doctor profile management and related features and collaborates only with the user profile module for patient-doctor communications and recommendations.

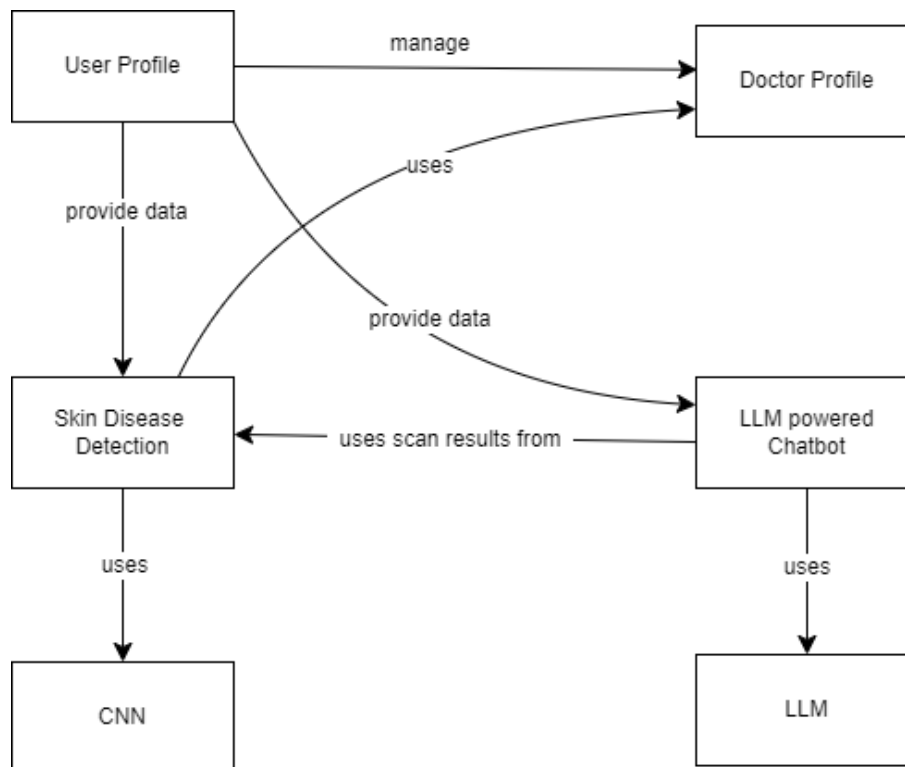


Figure 3.1: Modulor Relationship

3.1.2 Architectural Pattern

The "client-server" architecture is used in the development of DermaScan app which is modularized as follows:

- **Client (Mobile App):** Includes user profile modules which allows user to access the app and post skin scans for disease detection.
- **Server:** The server side is responsible for the functioning of CNN and LLM models in addition to results and reports generation based on user input.

A **Layered (Multi-tiered)** Architecture will be adopted.

1. **Presentation Layer** - User Interface.
2. **Application Layer** - Handles app logic, chatbot interaction, and model training.
3. **Data Layer** - Manage patient's Data, Doctors Information, manage medical records.

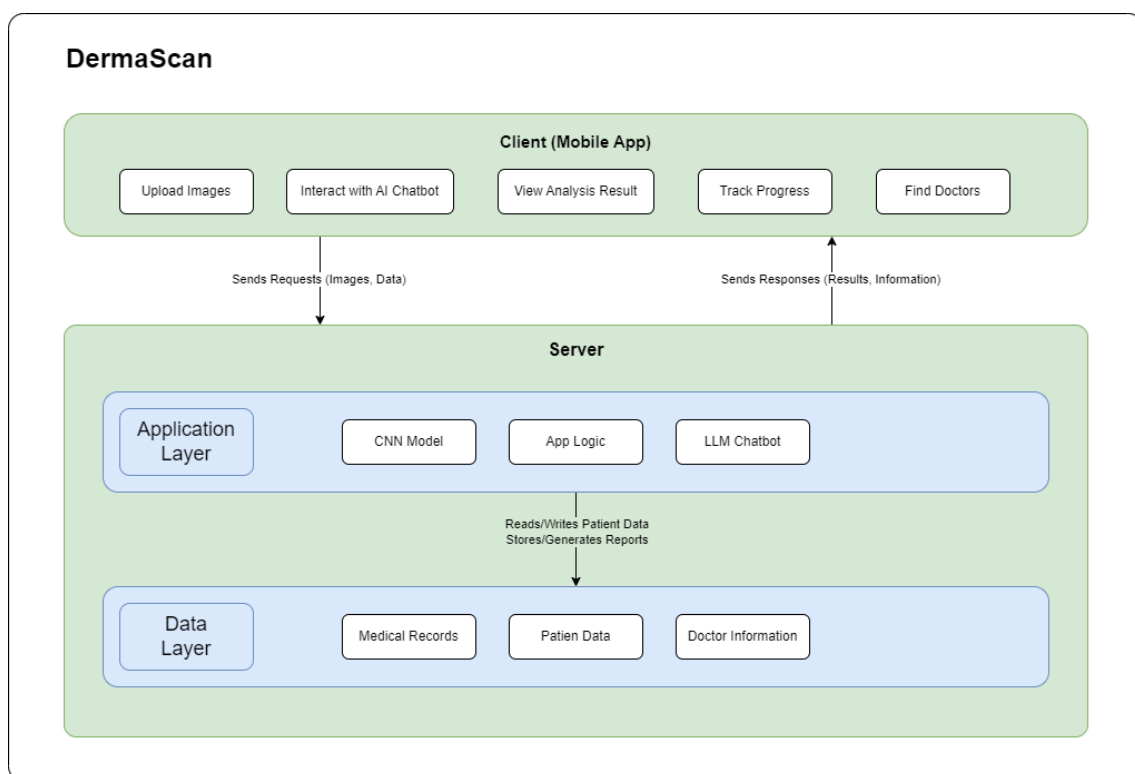


Figure 3.2: Architectural Pattern

3.2 Design Models

In this section, we illustrate various design models that are applicable to the system to ensure clarity in understanding the system architecture.

3.2.1 Activity Diagram

Following are the activity diagram representing the flow of activities and actions within the system.

3.2.1.1 Use Case 1: Register User

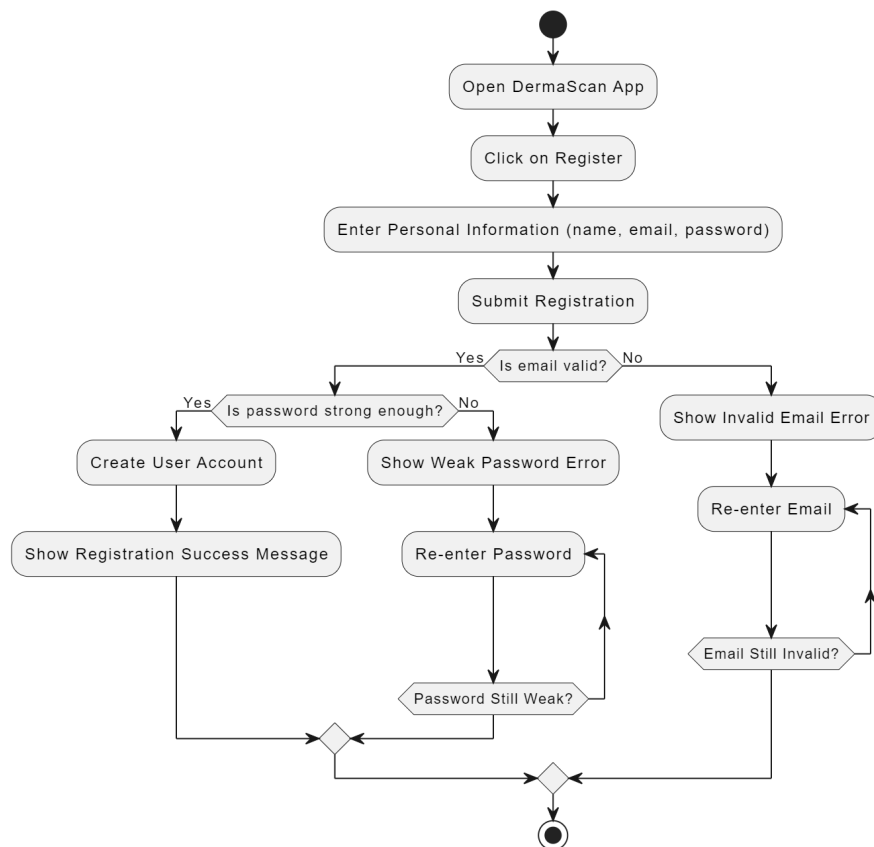


Figure 3.3: Activity Diagram Register User

3.2.1.2 Use Case 2: User Login

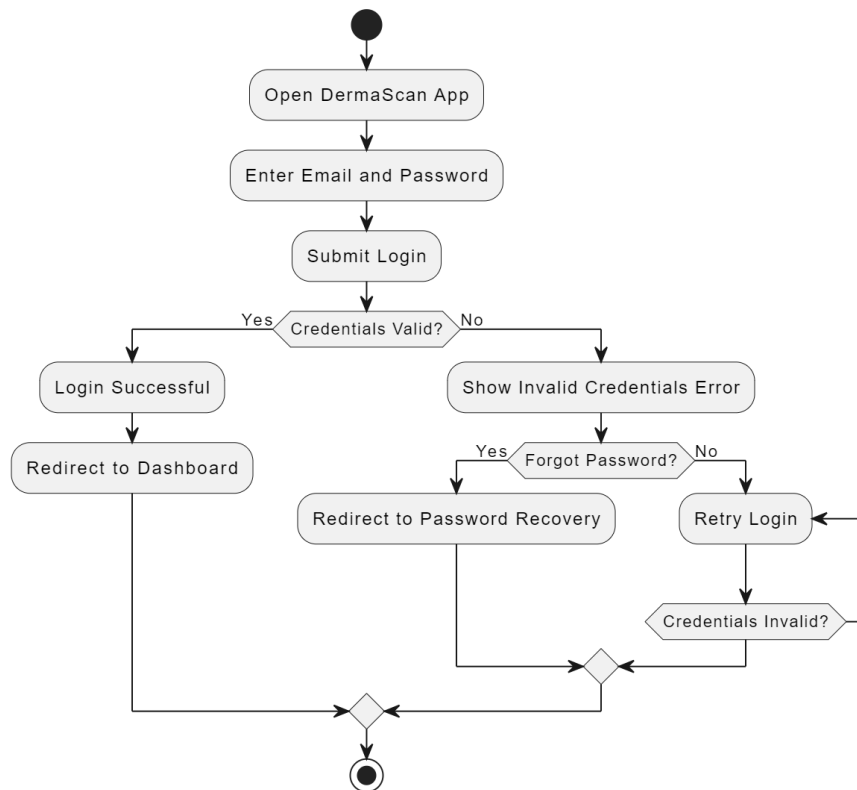


Figure 3.4: Activity Diagram User Login

3.2.1.3 Use Case 3: Manage User Profile

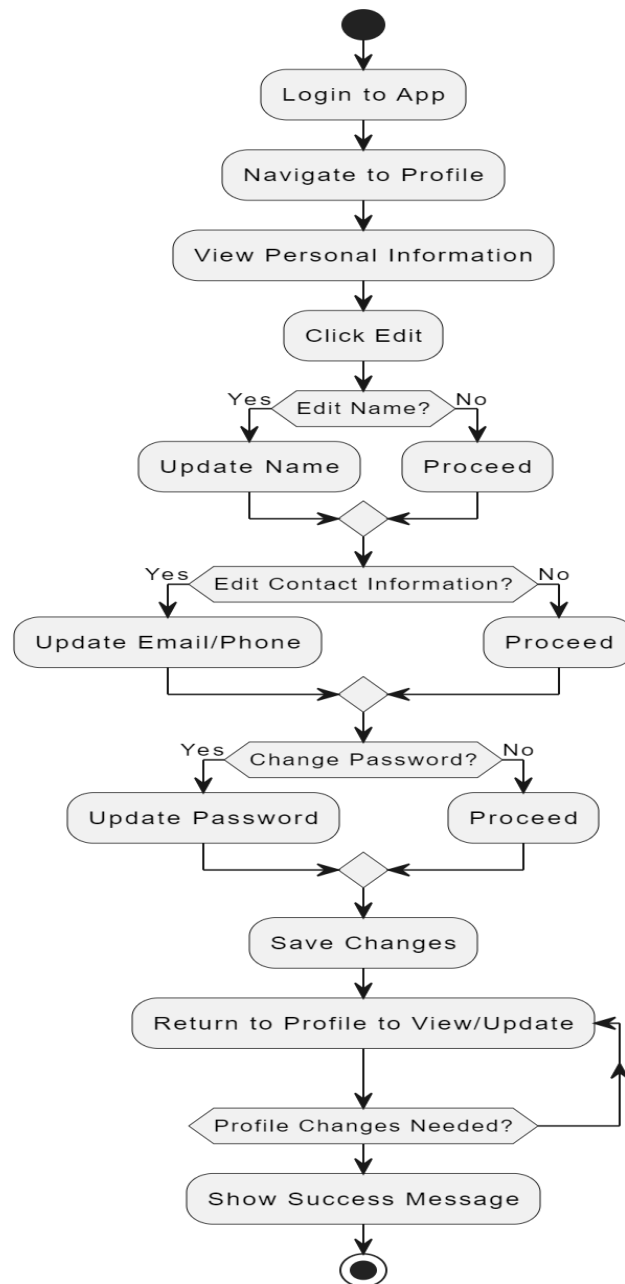


Figure 3.5: Activity Diagram Manage User Profile

3.2.1.4 Use Case 4: Perform Skin Scan

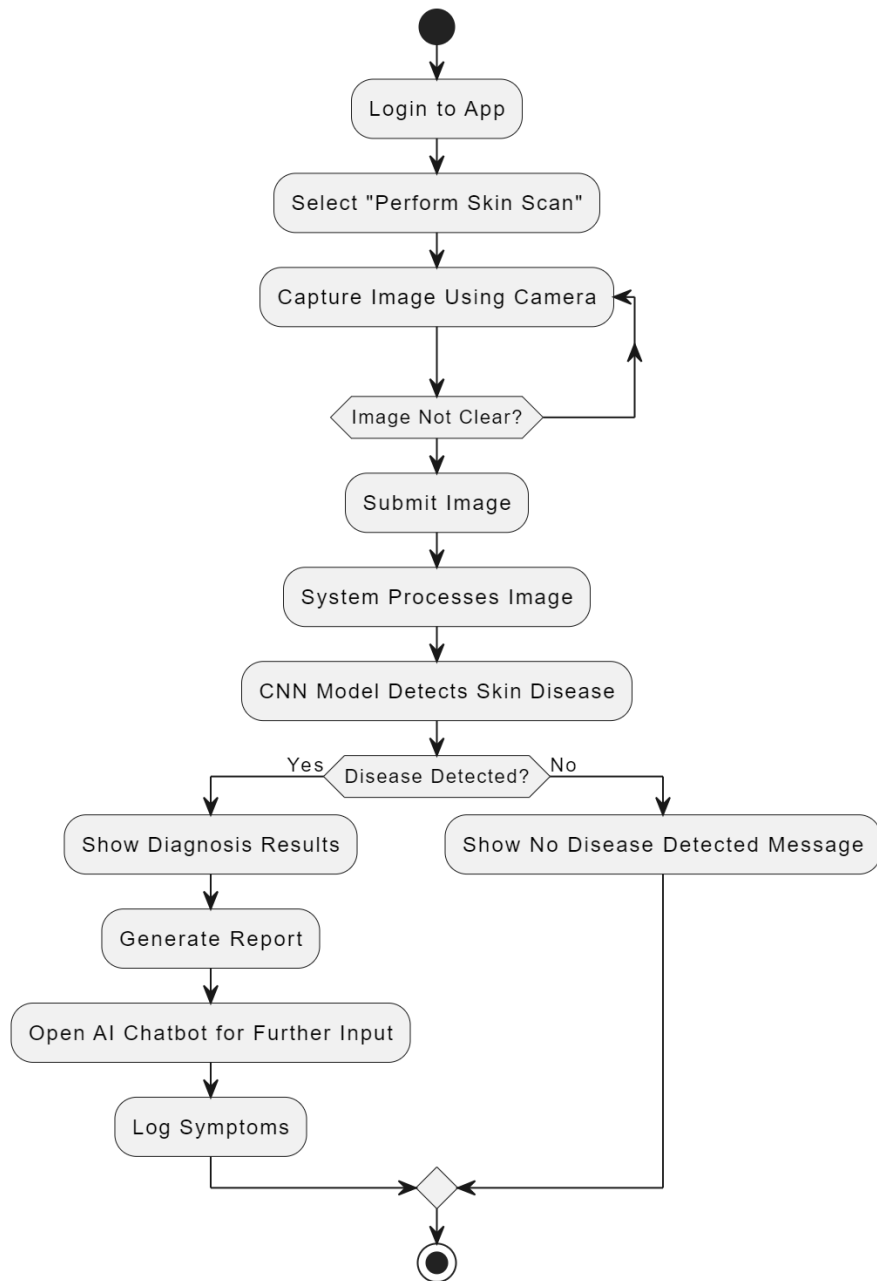


Figure 3.6: Activity Diagram Perform Skin Scan

3.2.1.5 Use Case 5: View Skin Analysis Result

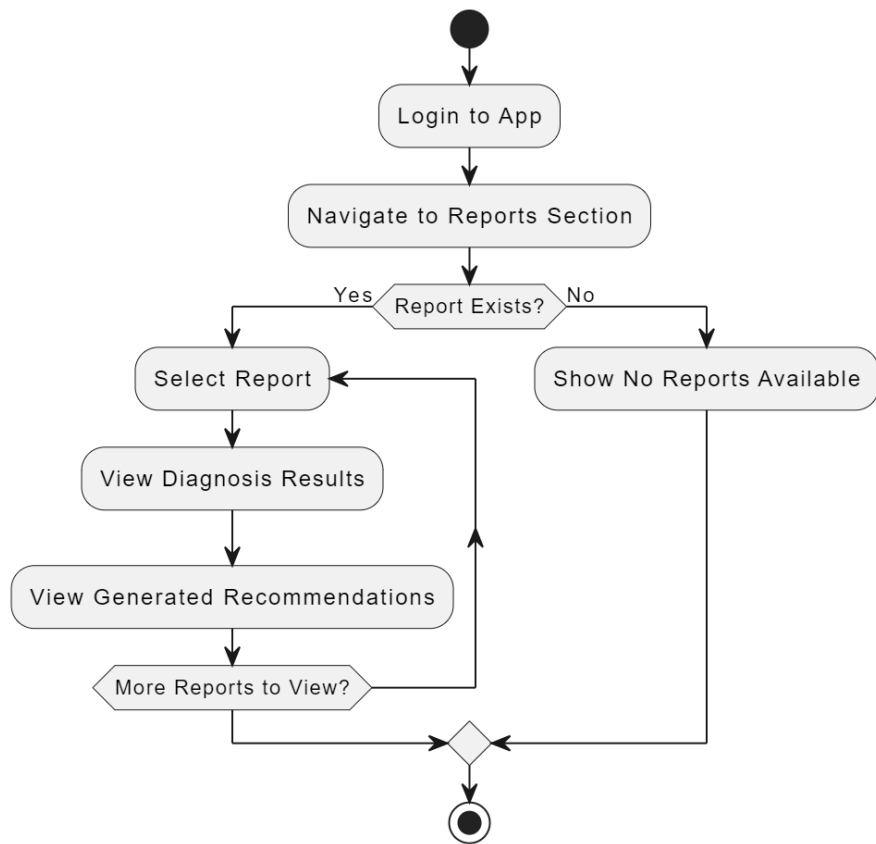


Figure 3.7: Activity Diagram View Skin Analysis Result

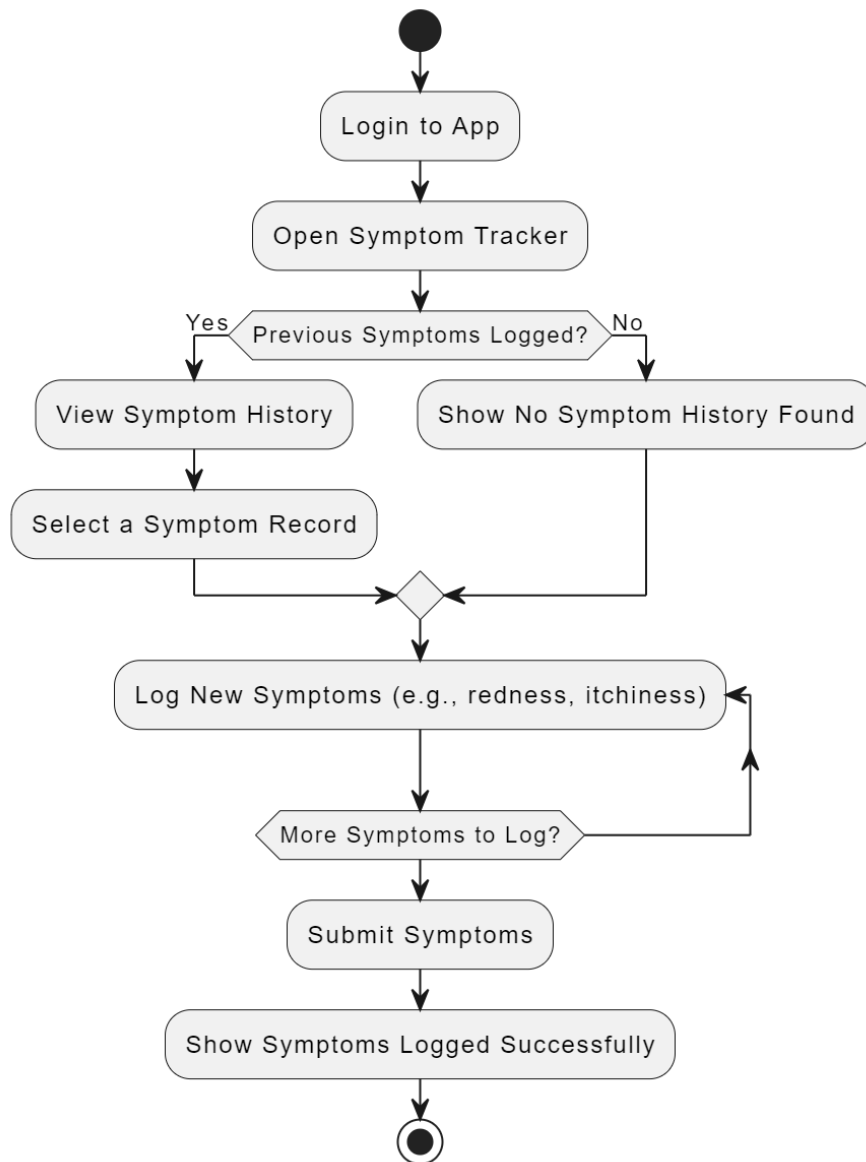
3.2.1.6 Use Case 6: Track Symptoms

Figure 3.8: Activity Diagram Track Symptoms

3.2.1.7 Use Case 7: Chatbot Interaction for Skin Advice

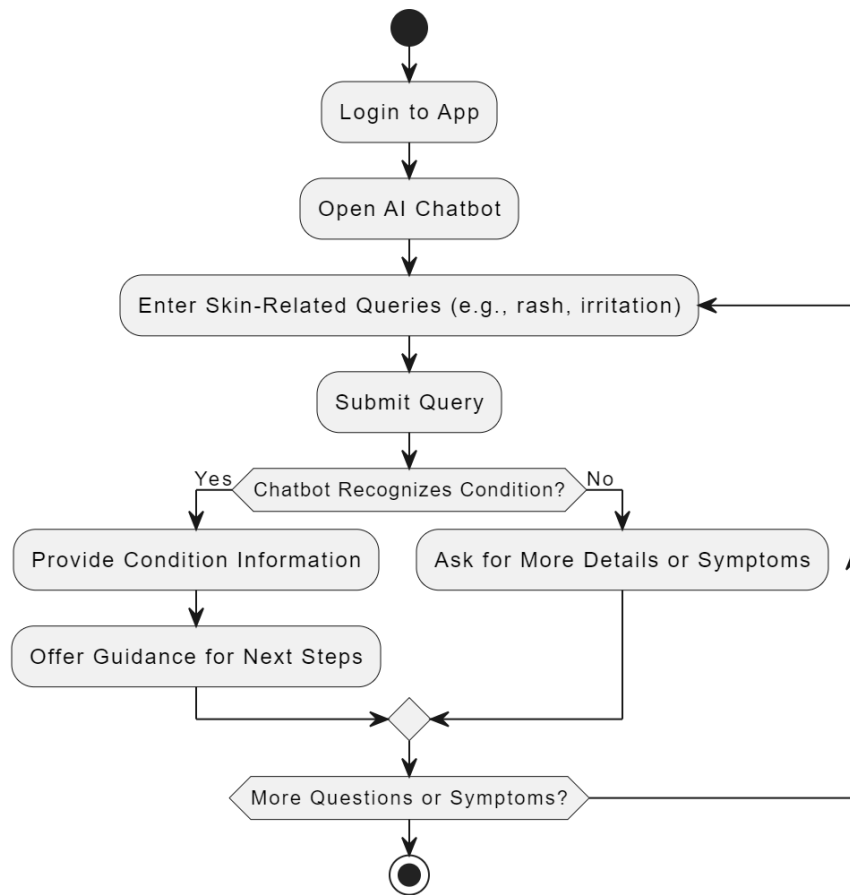


Figure 3.9: Activity Diagram Chatbot Interaction for Skin Advice

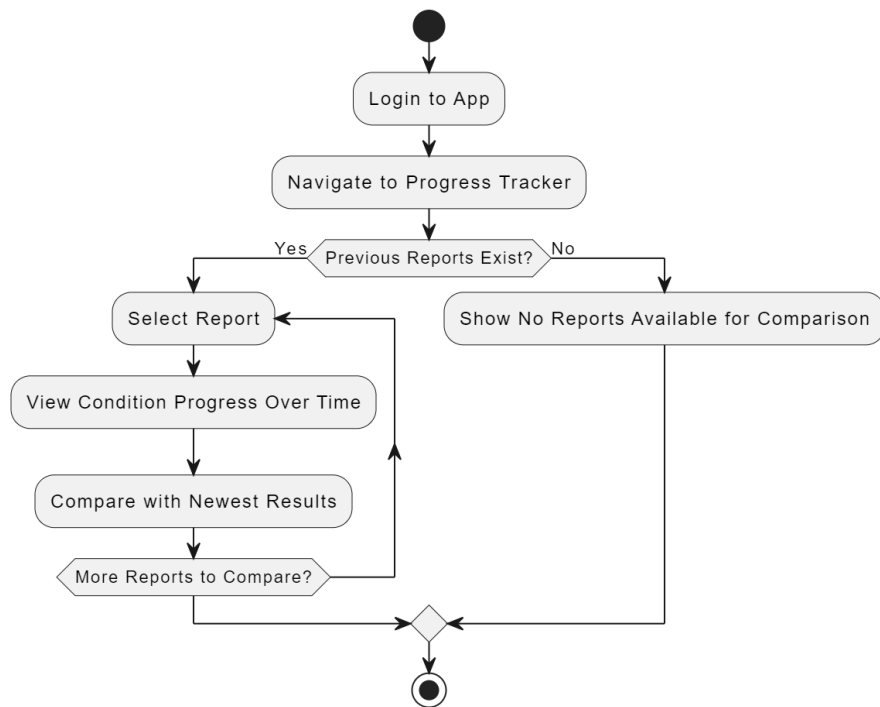
3.2.1.8 Use Case 8: Monitor Skin Condition Progress

Figure 3.10: Activity Diagram Monitor Skin Condition Progress

3.2.1.9 Use Case 9: Manage Doctor Recommendations

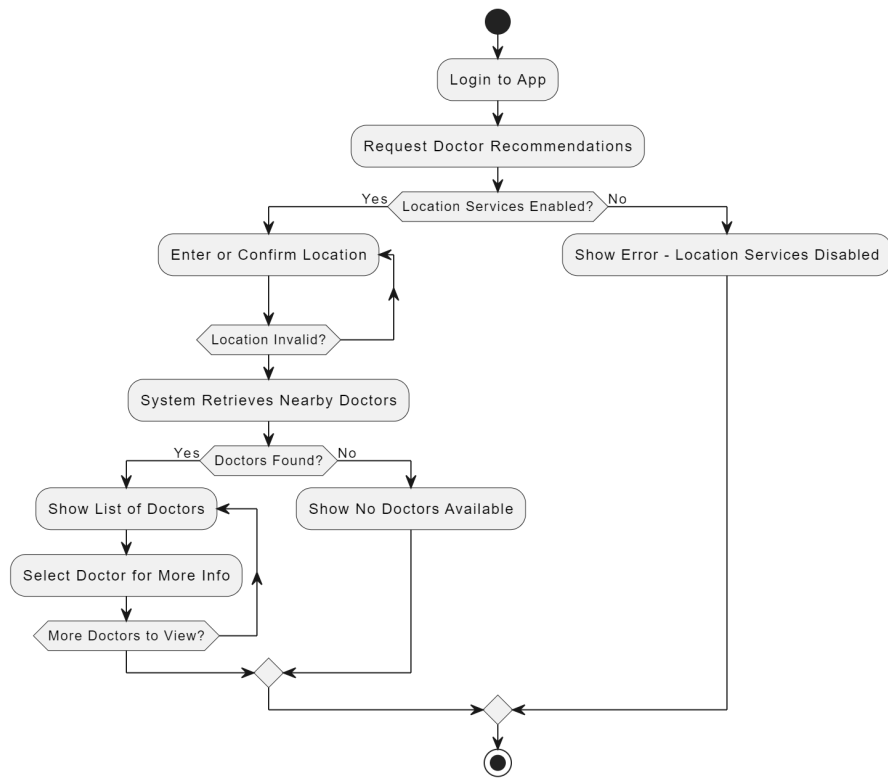


Figure 3.11: Activity Diagram Manage Doctor Recommendations

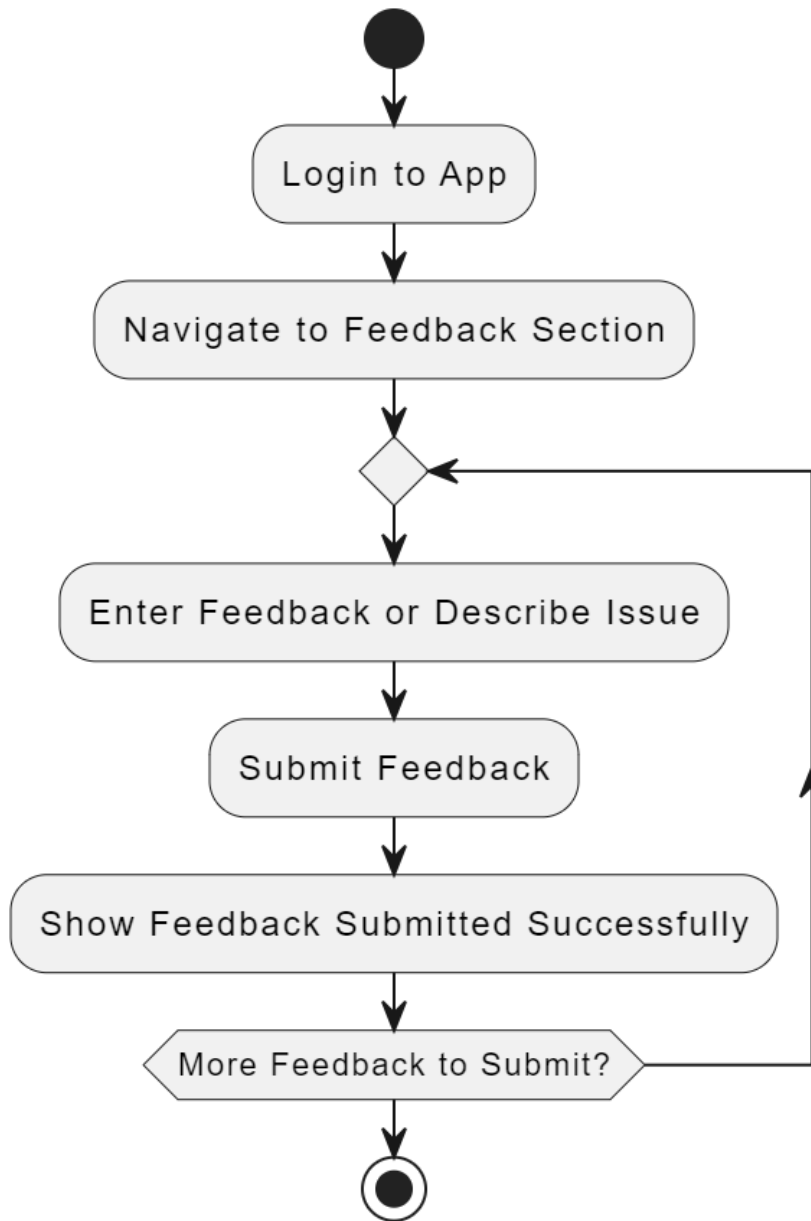
3.2.1.10 Use Case 10: Provide Feedback and Report Issues

Figure 3.12: Activity Diagram Provide Feedback and Report Issues

3.2.1.11 Use Case 11: View Dermatologist Profiles

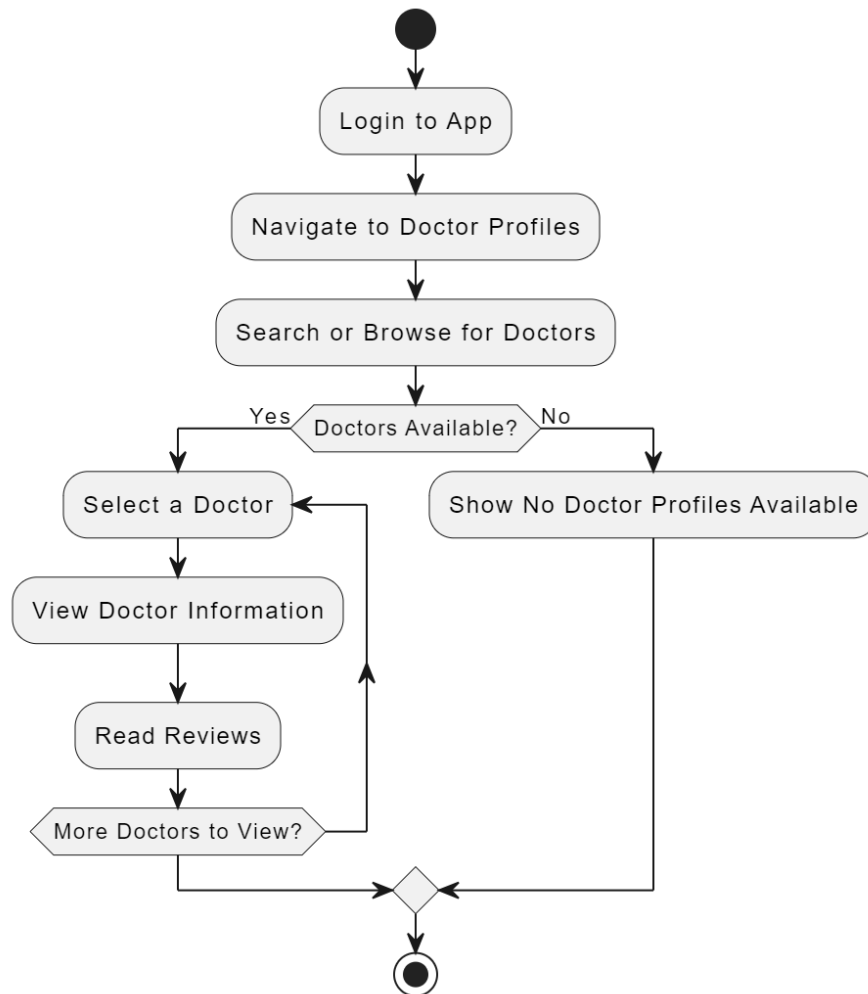


Figure 3.13: Activity Diagram View Dermatologist Profiles

3.2.1.12 Activity Diagram (Whole System)

Following is the Activity Diagram of the entire system illustrating how different components interact and work together.

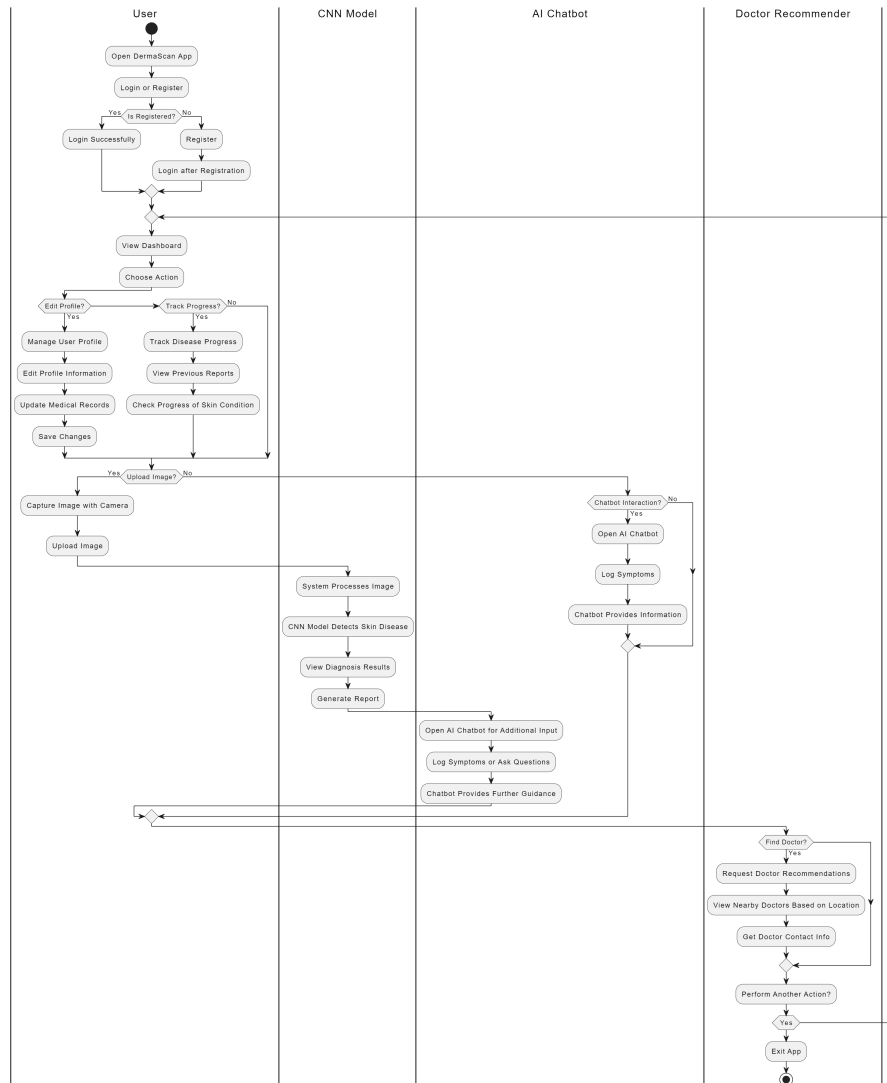


Figure 3.14: Activity Diagram Whole System

3.2.2 Class Diagram

The Class Diagram below illustrates the key entities, their attributes, operations, and the relationships between them.

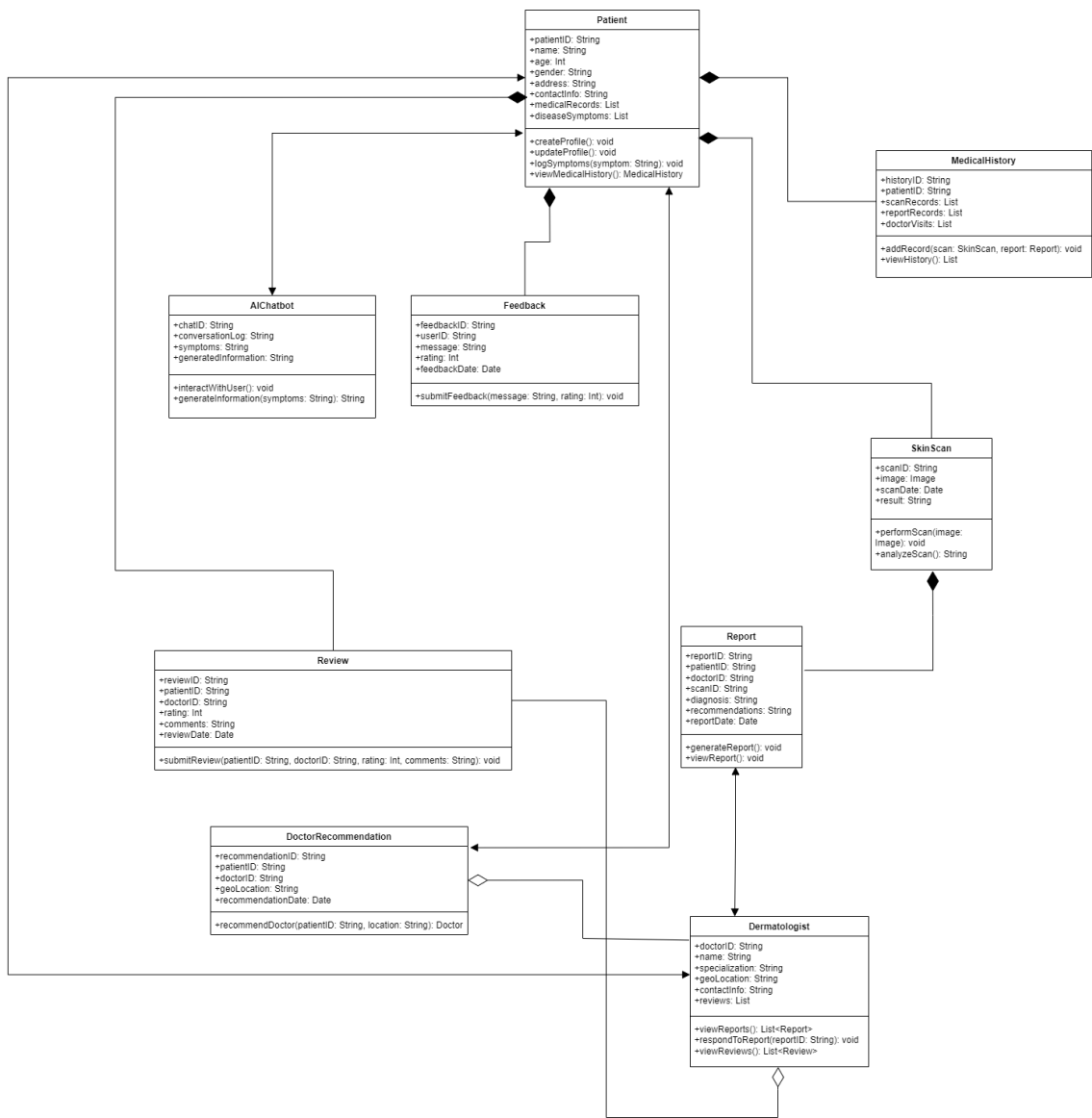


Figure 3.15: Class Diagram

3.2.3 Class-level Sequence Diagram

3.2.3.1 Use Case 1: Register User

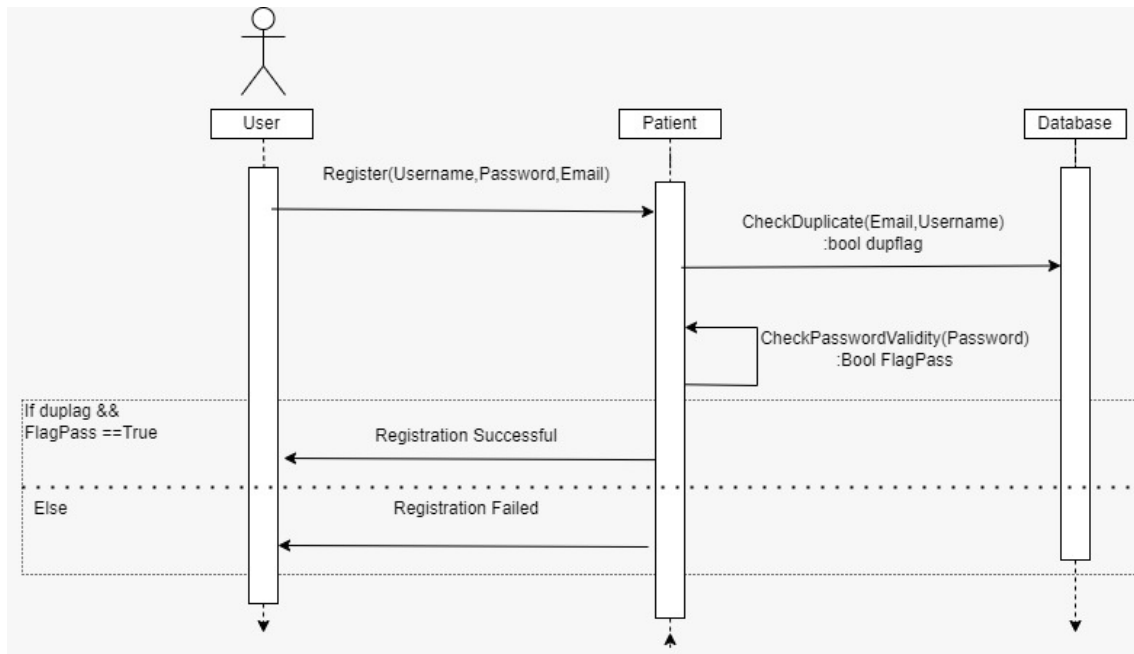


Figure 3.16: Sequence Diagram Register User

3.2.3.2 Use Case 2: User Login

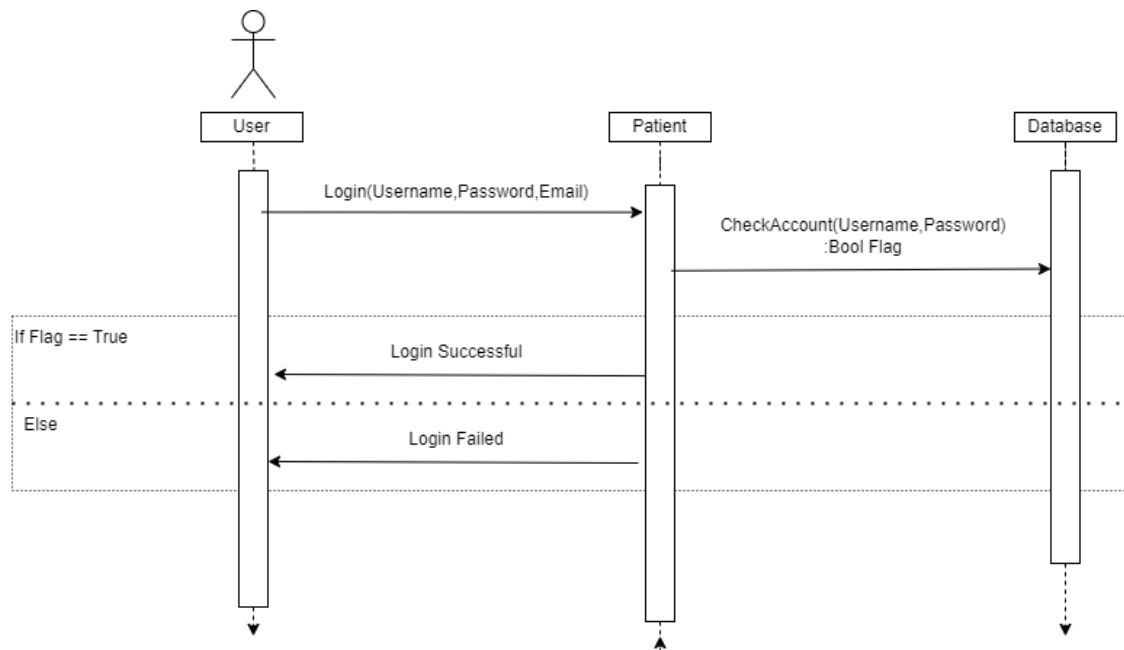


Figure 3.17: Sequence Diagram User Login

3.2.3.3 Use Case 3: Manage User Profile

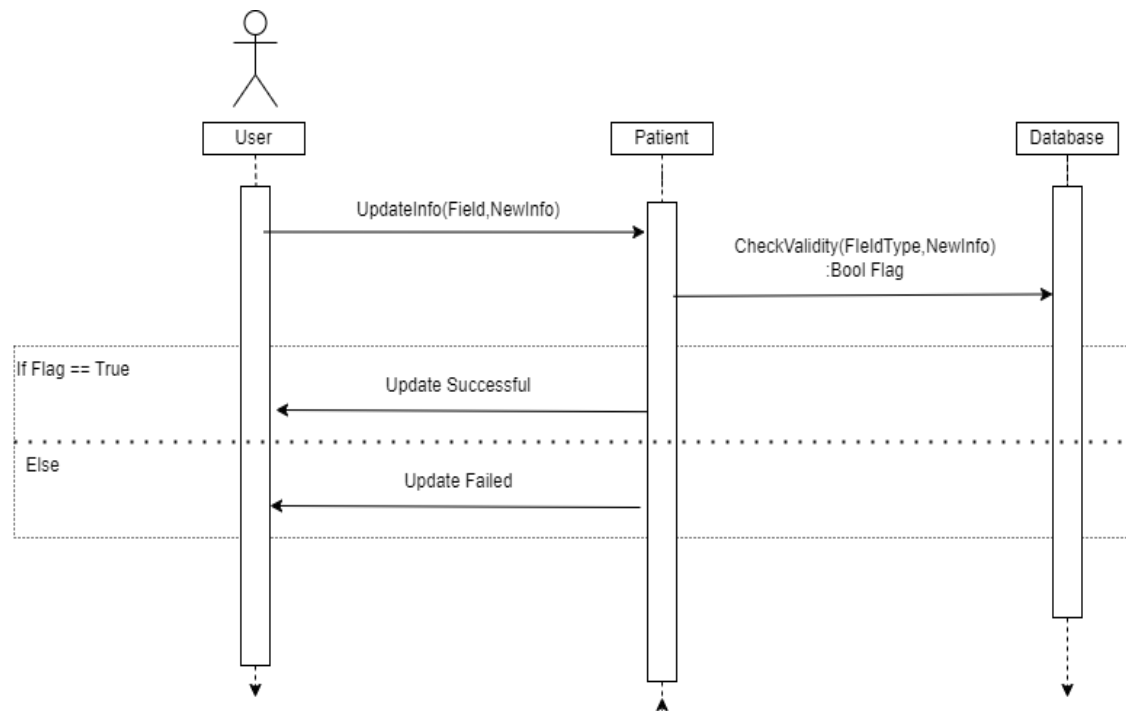


Figure 3.18: Sequence Diagram Manage User Profile

3.2.3.4 Use Case 4: Perform Skin Scan

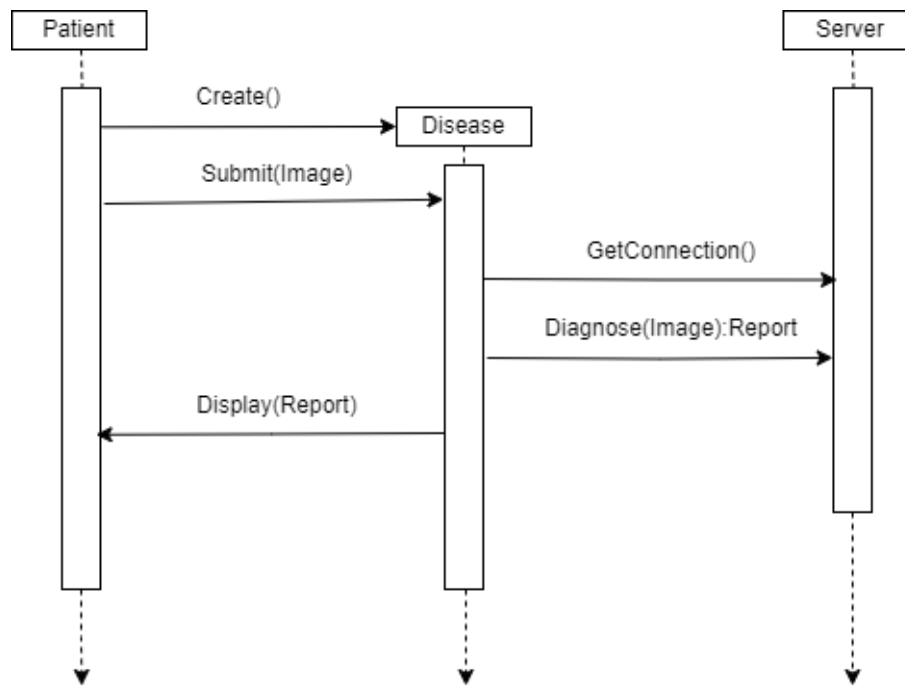


Figure 3.19: Sequence Diagram Perform Skin Scan

3.2.3.5 Use Case 5: View Skin Analysis Result

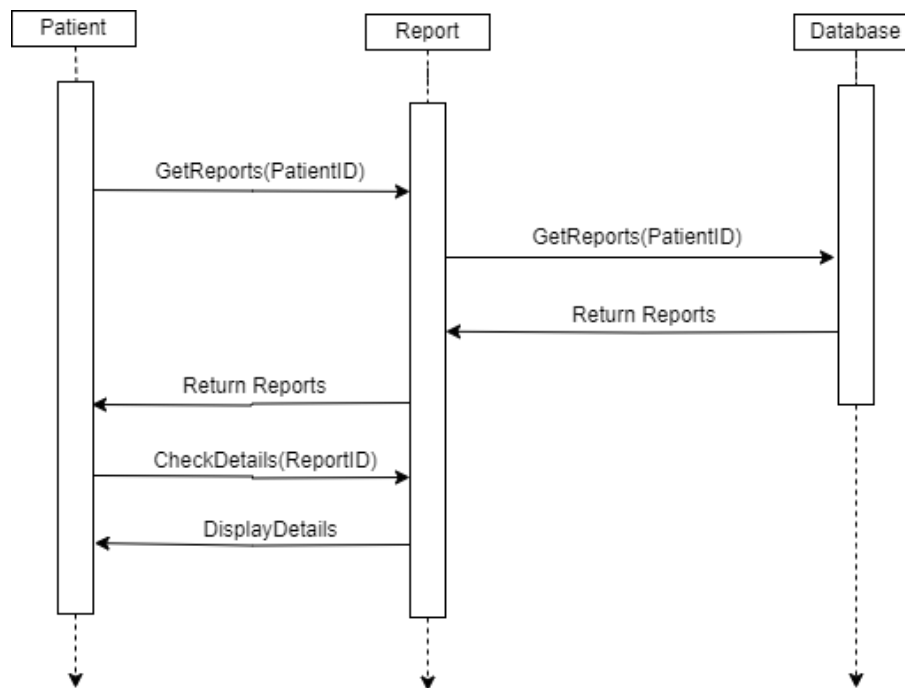


Figure 3.20: Sequence Diagram View Skin Analysis Result

3.2.3.6 Use Case 6: Track Symptoms

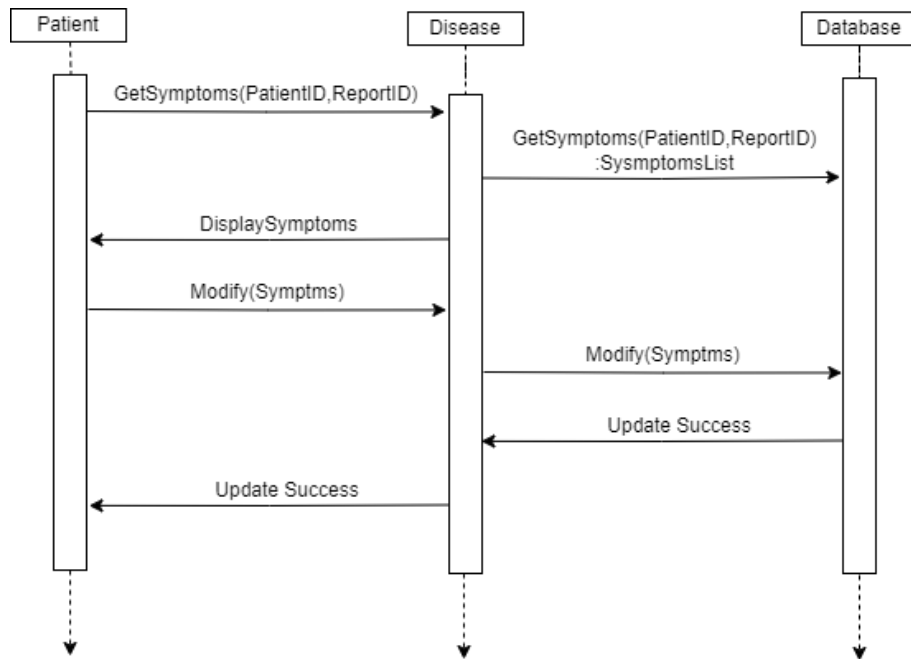


Figure 3.21: Sequence Diagram Track Symptoms

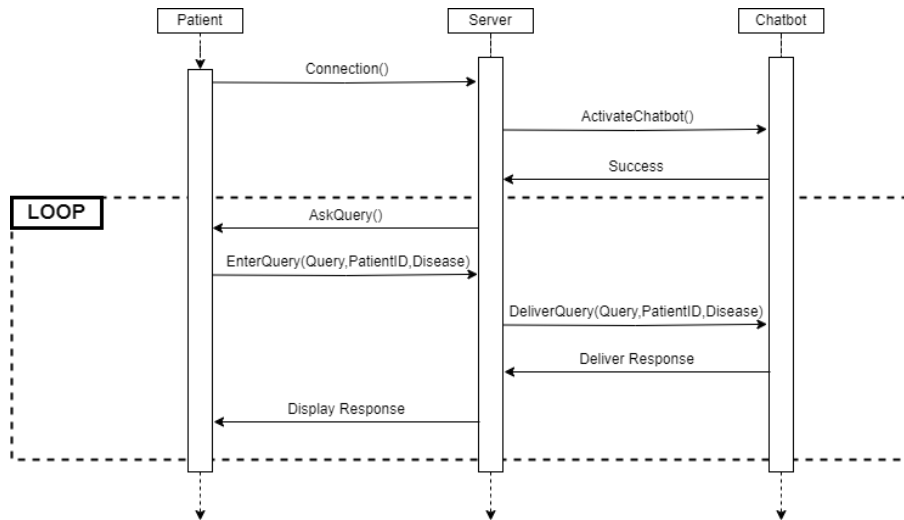
3.2.3.7 Use Case 7: Chatbot Interaction for Skin Advice

Figure 3.22: Sequence Diagram Chatbot Interaction for Skin Advice

3.2.3.8 Use Case 8: Monitor Skin Condition Progress

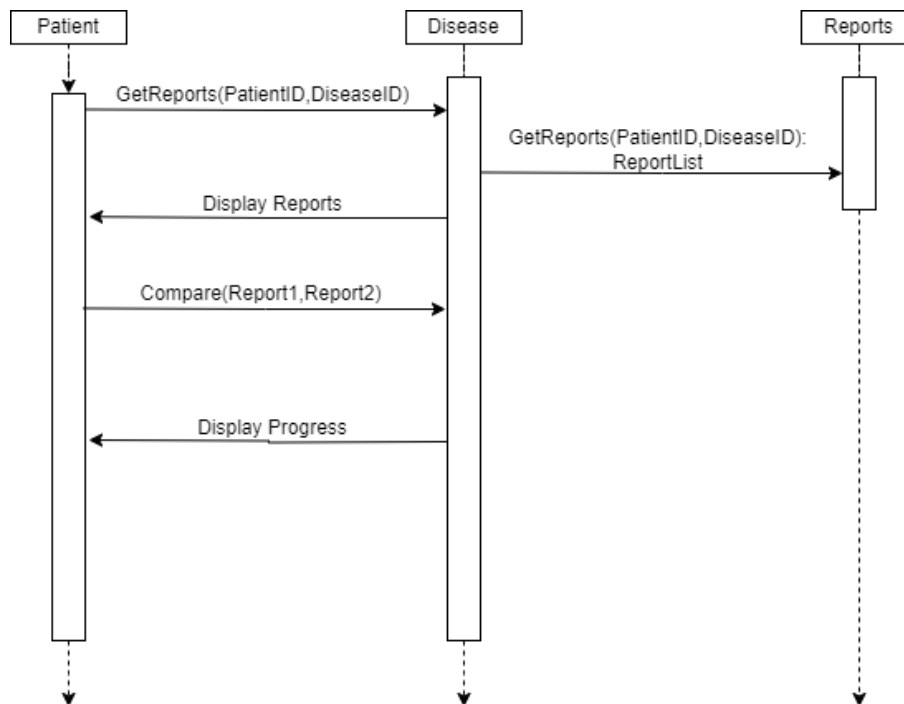


Figure 3.23: Sequence Diagram Monitor Skin Condition Progress

3.2.3.9 Use Case 9: Manage Doctor Recommendations

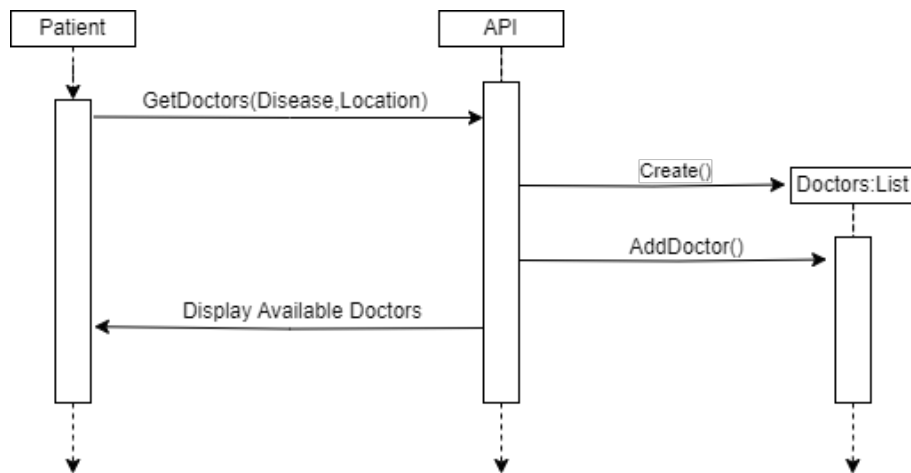


Figure 3.24: Sequence Diagram Manage Doctor Recommendations

3.2.3.10 Use Case 10: Provide Feedback and Report Issues

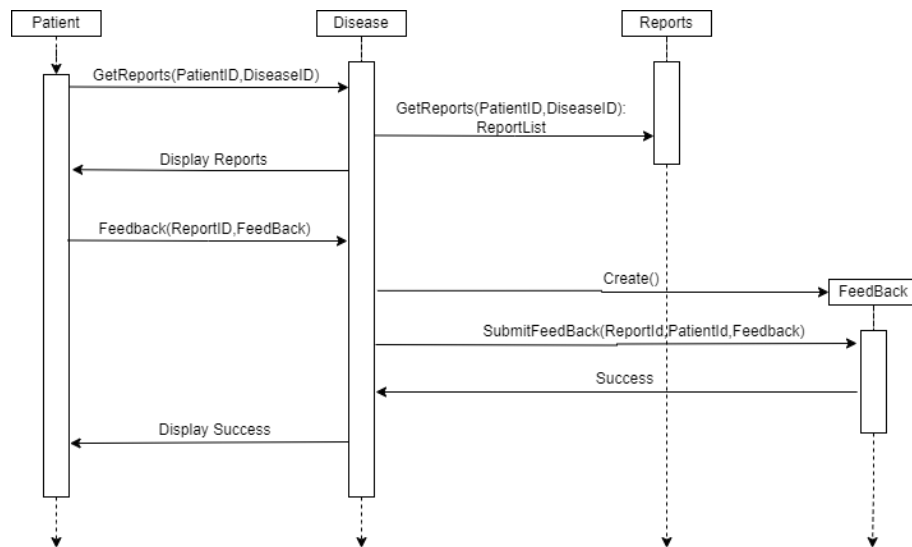


Figure 3.25: Sequence Diagram Provide Feedback and Report Issues

3.2.3.11 Use Case 11: View Dermatologist Profiles

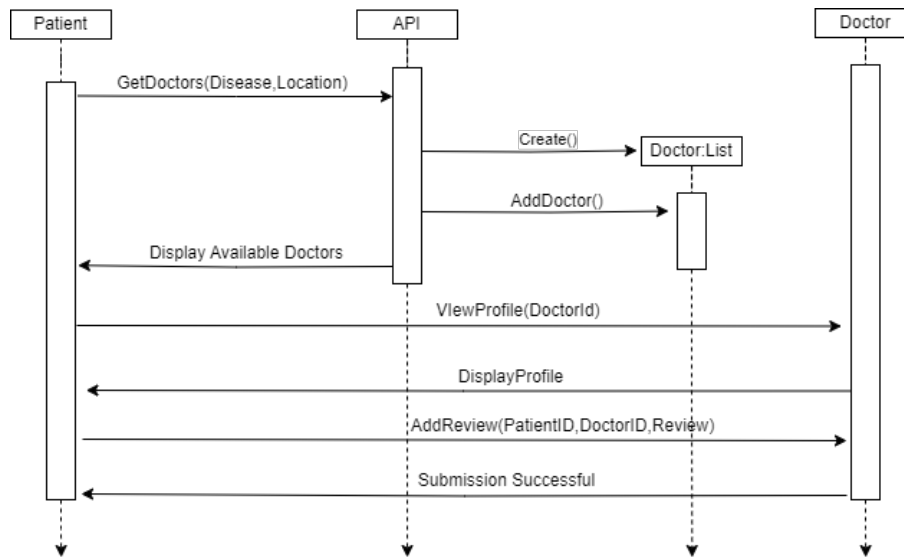


Figure 3.26: Sequence Diagram View Dermatologist Profiles

3.2.4 State Transition Diagram

Following are the State transition diagrams of the product.

1. User Profile

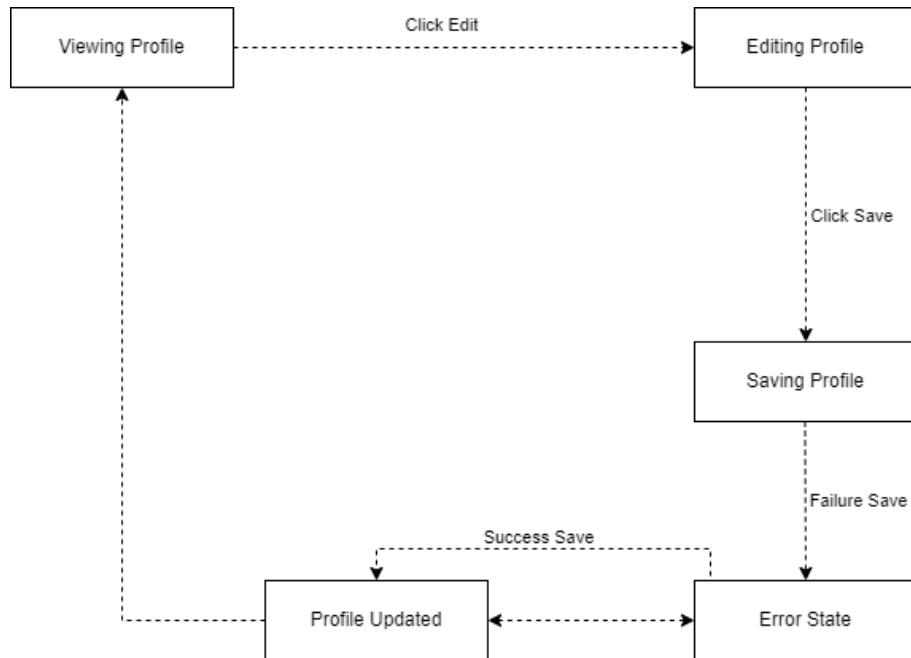


Figure 3.27: State Transition User Profile

2. Skin Disease Scanning Process

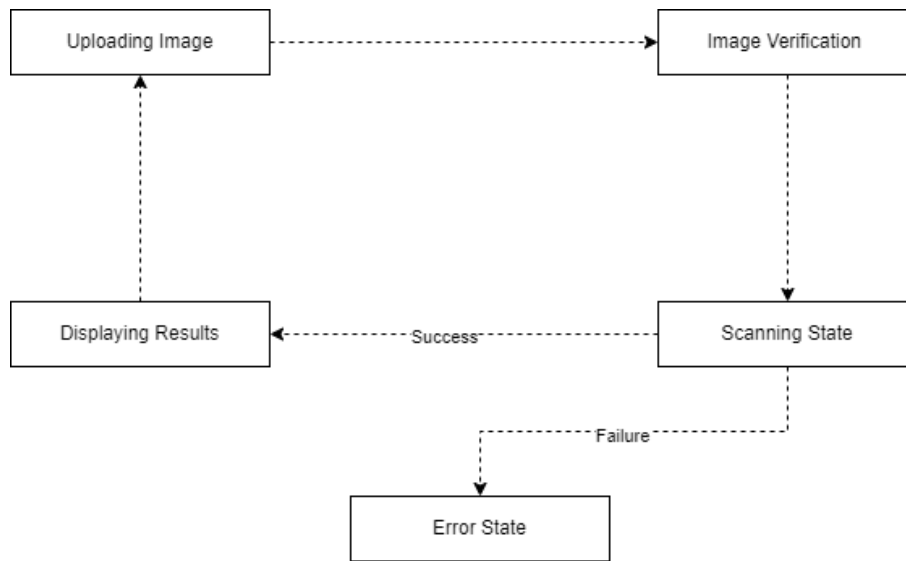


Figure 3.28: State Transition Skin Scanning Process

3. Chatbot Interaction for Skin Advice

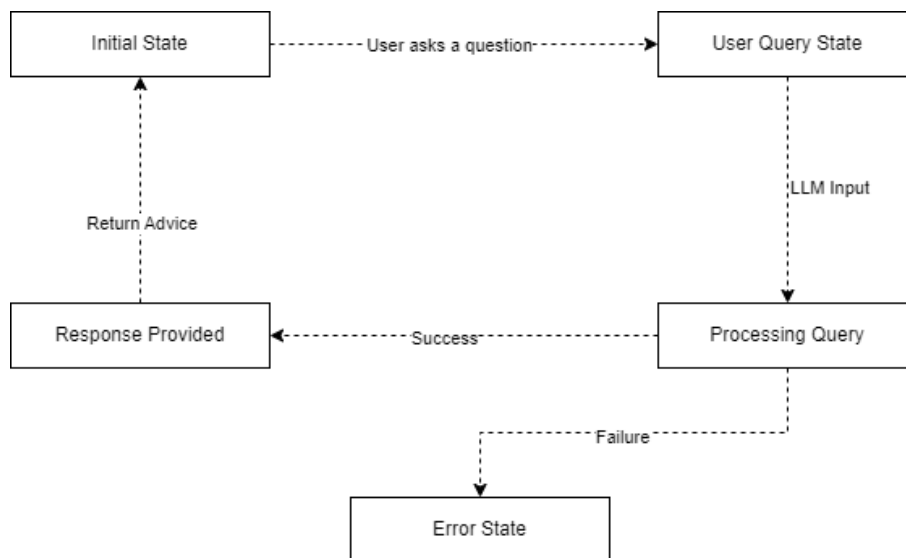


Figure 3.29: State Transition Chatbot Interaction

4. Tracking Skin Condition Progress

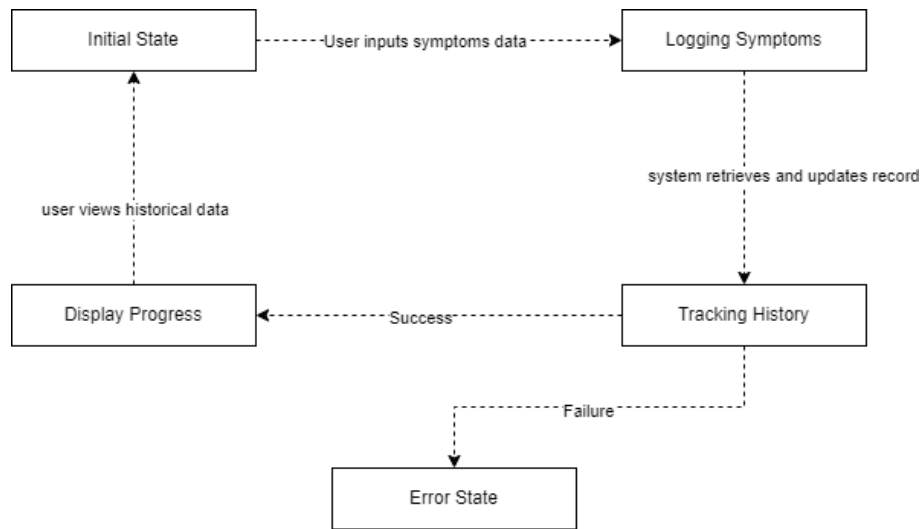


Figure 3.30: State Transition Tracking Skin Condition Progress

3.3 Data Design

DermaScan, being an image-processing application consists of data-intensive features for the accurate detection and diagnosis of skin-diseases including the data storage of datasets for training the CNN and LLM models. The information domain of the app contains multiple entities including patients, doctors and tables for data storage of chatbot conversations, maintaining medical records and image storage for skin-disease detection. TO achieve efficiency in data storage, processing and retrieval, our app uses the Firebase as its primary database to store all data relevant to the application. A detailed explanation and a visual representation of DermaScan's data transformation into data structures is given below.

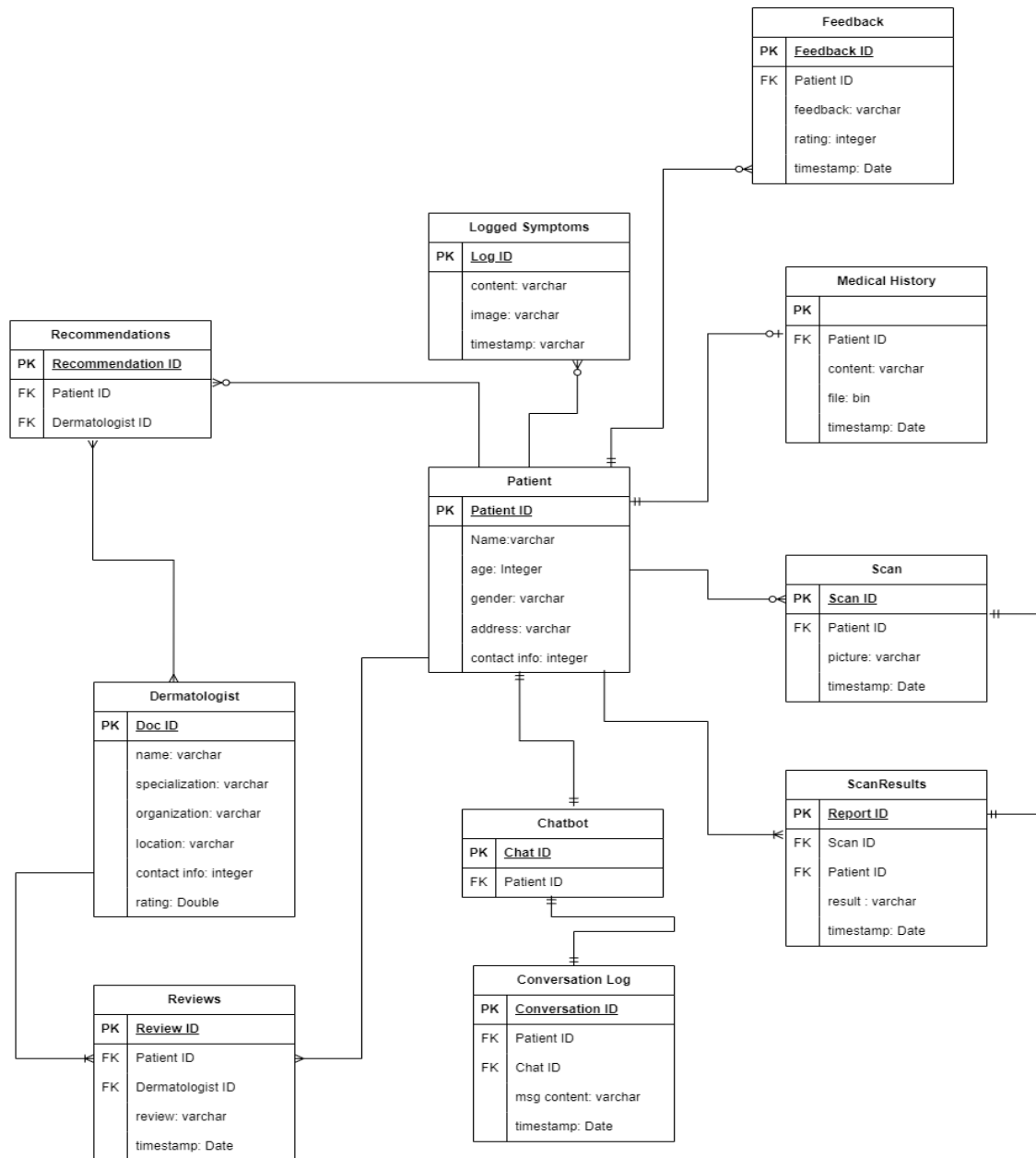


Figure 3.31: ERD

3.3.1 Data Structures

- **Patient**
 - **Attributes:** PatientID(PK), name, age, gender, address, contact
 - **Description:** Stores users basic information
- **Medical History**
 - **Attributes:** PatientID(FK), content, file, timestamp
 - **Description:** Stores users medical history
- **Scan**
 - **Attributes:** ScanID(PK), PatientID(FK), picture, timestamp
 - **Description:** Stores users input for diagnostic scan
- **Scan results**
 - **Attributes:** ReportID(PK), ScanID(FK), PatientID(FK), result, timestamp
 - **Description:** stores users previous diagnostic scan results
- **Logged symptoms**
 - **Attributes:** LogID(PK), content, image, timestamp
 - **Description:** Stores users symptoms with respect to scans.
- **Chatbot**
 - **Attributes:** ChatID(PK), PatientID(FK)
 - **Description:** Stores users chatbot information
- **Conversation log**
 - **Attributes:** PatientID(FK), ChatID(FK), message, timestamp
 - **Description:** Stores conversation between chatbot and user.
- **Recommendations**
 - **Attributes:** RecommendationID(PK), PatientID(FK), DermatologistID(FK)
 - **Description:** Stores recommendations provided to user.

- **Dermatologist**

- **Attributes:** DocID(PK), name, specialization, organization, location, contactInfo, rating
- **Description:** Stores information about dermatologists.

- **Review**

- **Attributes:** ReviewID(PK), PatientID(FK), DocID(FK), review, timestamp
- **Description:** Stores reviews posted by users towards dermatologists.

- **Feedback**

- **Attributes:** FeedbackID(PK), PatientID(FK), feedback, timestamp, rating
- **Description:** Stores feedback given by users on their diagnostics.

Database:

Firestore - DermaScan, as a skin disease detection app, heavily relies on a CNN model, that works on images for disease detection. In this regard, the most optimal database for DermaScan is Firestore, which offers a comprehensive suite of services, including secure storage, authentication, and real-time database management.

Bibliography